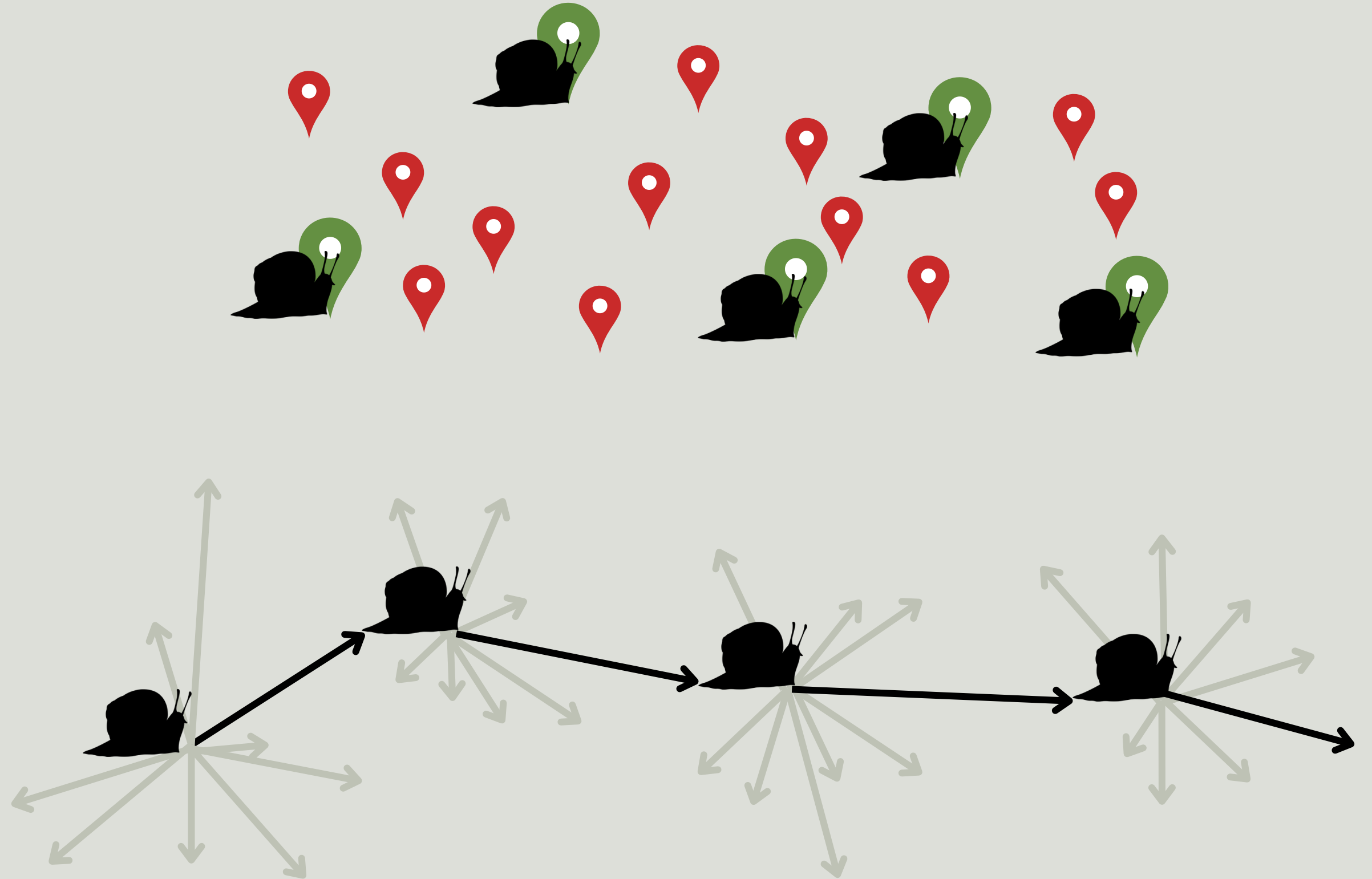


Selección

Recursos

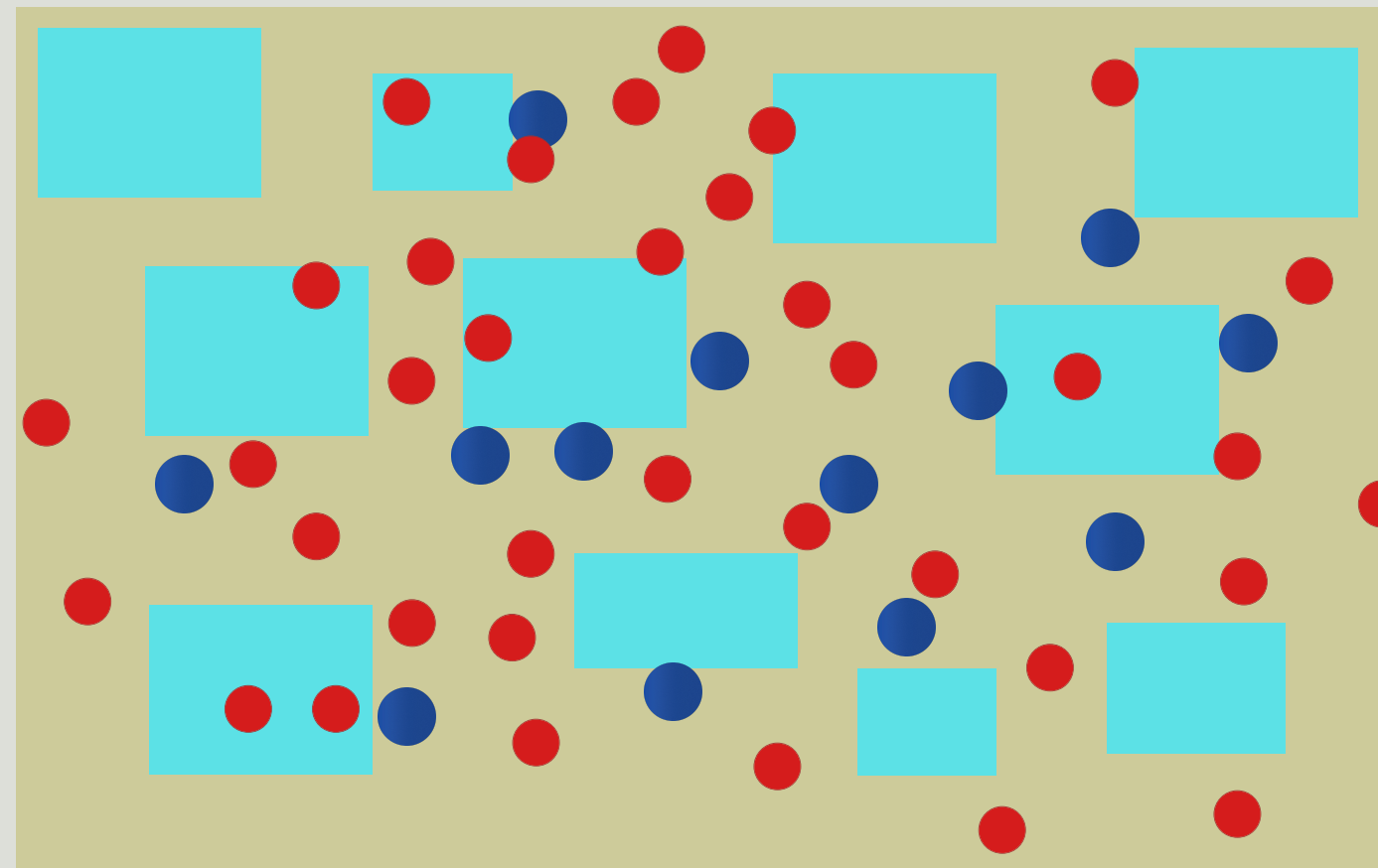
Pasos



Resource Selection Functions — RSF

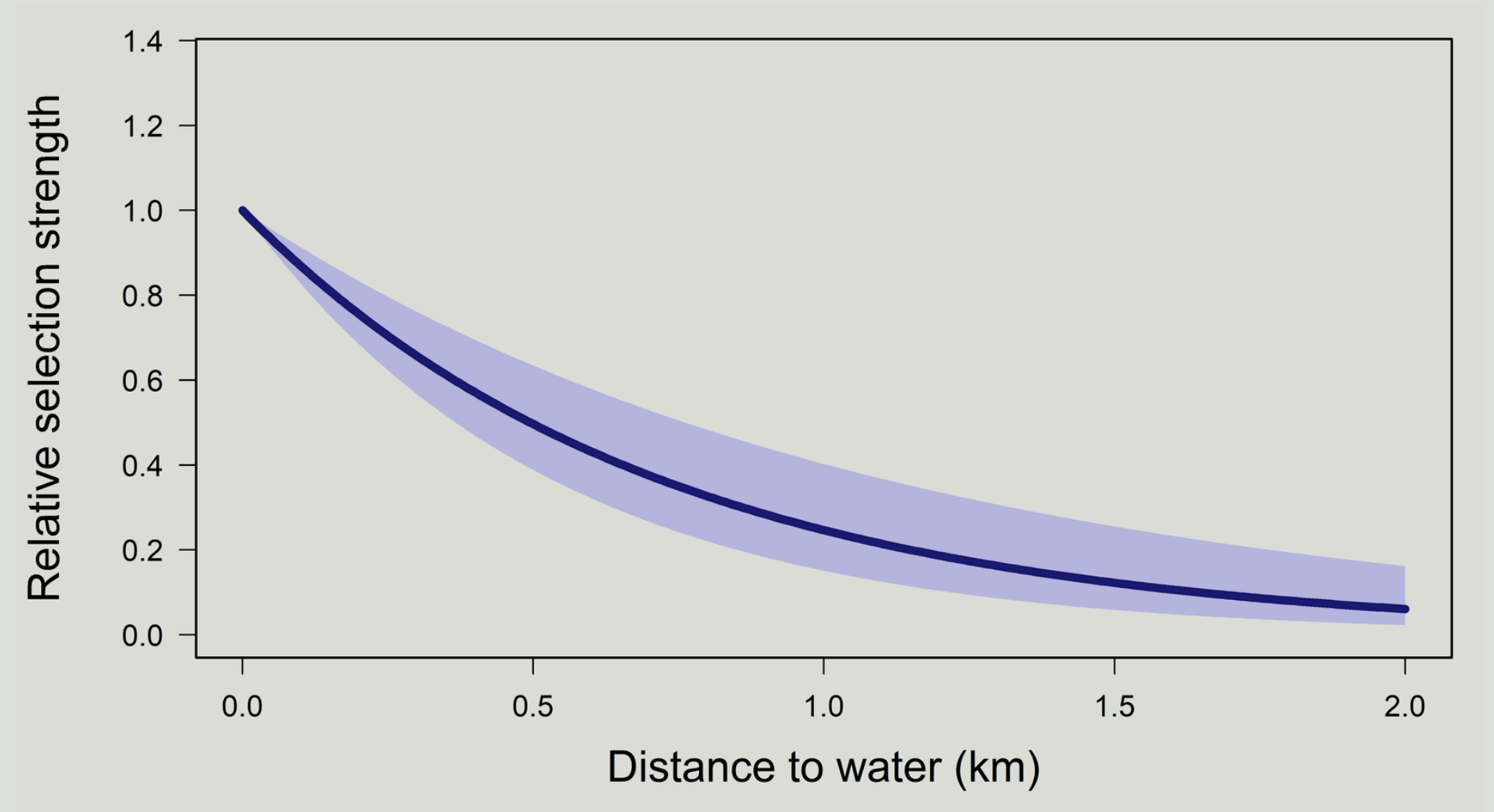
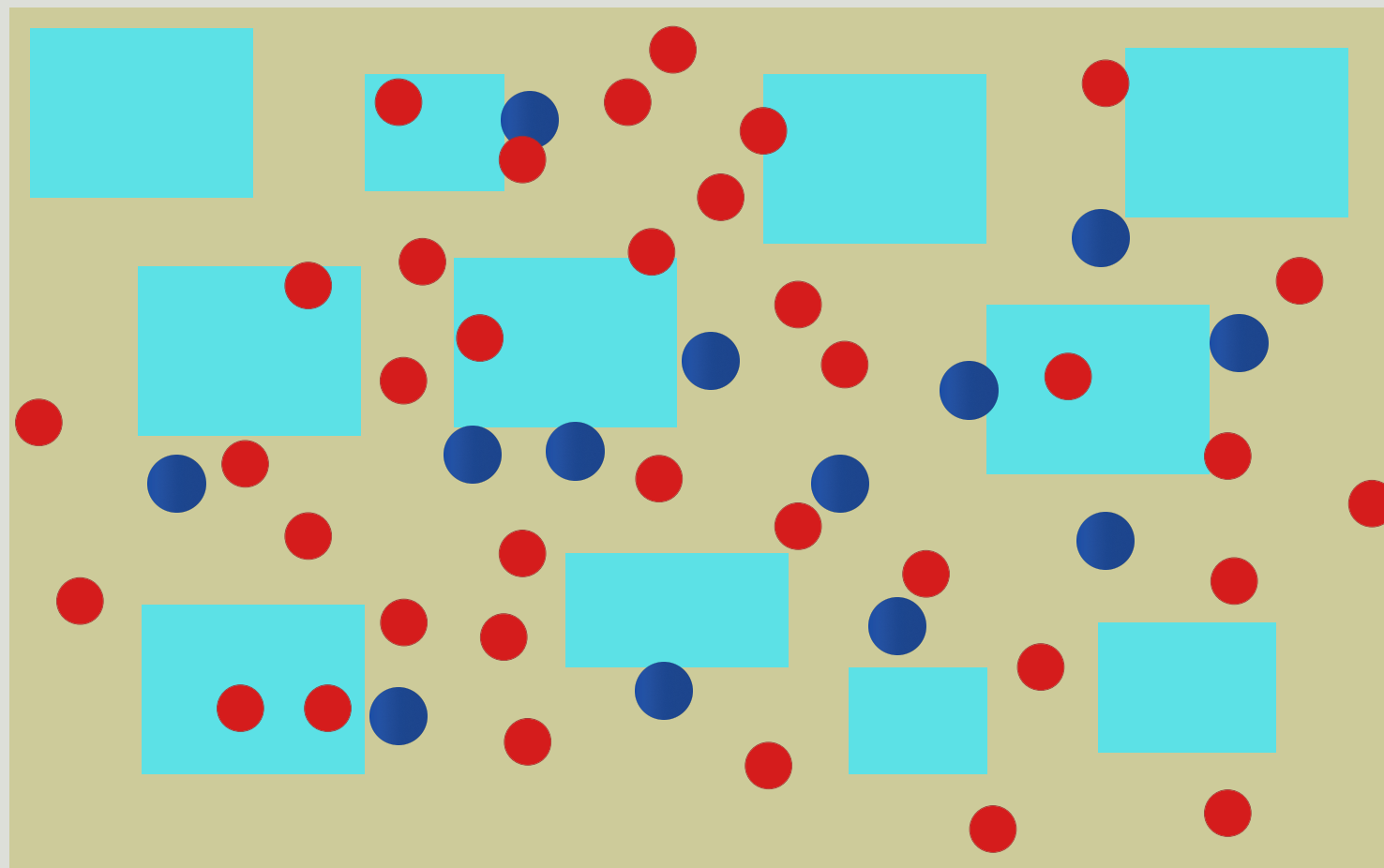
$$\text{Selección} \propto \frac{\text{disponibilidad}}{\text{uso}}$$

- Usado
- Disponible



Resource Selection Functions — RSF

$$\text{Selección} \propto \frac{\text{disponibilidad}}{\text{uso}}$$



Resource Selection Functions — RSF

Modelo lineal generalizado con respuesta binaria

$$Y_i = \begin{cases} 1, & \text{si el punto } i \text{ fue usado} \\ 0, & \text{si el punto } i \text{ fue disponible} \end{cases}$$

$$\text{logit}(p_i) = \log \left(\frac{p_i}{1 - p_i} \right) = \beta_0 + \beta_1 x_{i1} + \beta_2 x_{i2} + \cdots + \beta_p x_{ip}$$

$$w(x_i) = \exp (\beta_1 \text{bosque}_i + \beta_2 \text{pendiente}_i + \beta_3 \text{dist_agua}_i)$$

Regresión logística

Verosimilitud

NDVI, dist. agua...

$$L(\beta) = \prod_{i=1}^n P(y_i \mid x_i; \beta)^{w_i}$$

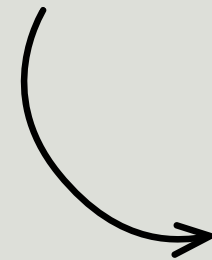
Usado o disponible

$$w_i = \begin{cases} w_1, & \text{si } y_i = 1 \\ w_0, & \text{si } y_i = 0 \end{cases}$$

The diagram illustrates the components of the logistic regression likelihood function. The term 'Verosimilitud' (Likelihood) points to the function $L(\beta)$. The input features, such as 'NDVI, dist. agua...', are represented by x_i . The output variable, 'Usado o disponible' (Used or available), is represented by y_i . The weights w_i are shown as exponents in the likelihood function and are defined by a piecewise function based on the output y_i .

Regresión logística

$$L(\beta) = \prod_{i=1}^n P(y_i \mid x_i; \beta)^{w_i}$$

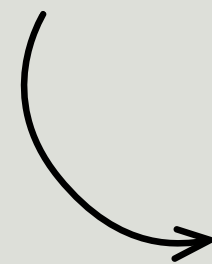


Observaciones
independientes



Regresión logística

$$L(\beta) = \prod_{i=1}^n P(y_i | x_i; \beta)^{w_i}$$



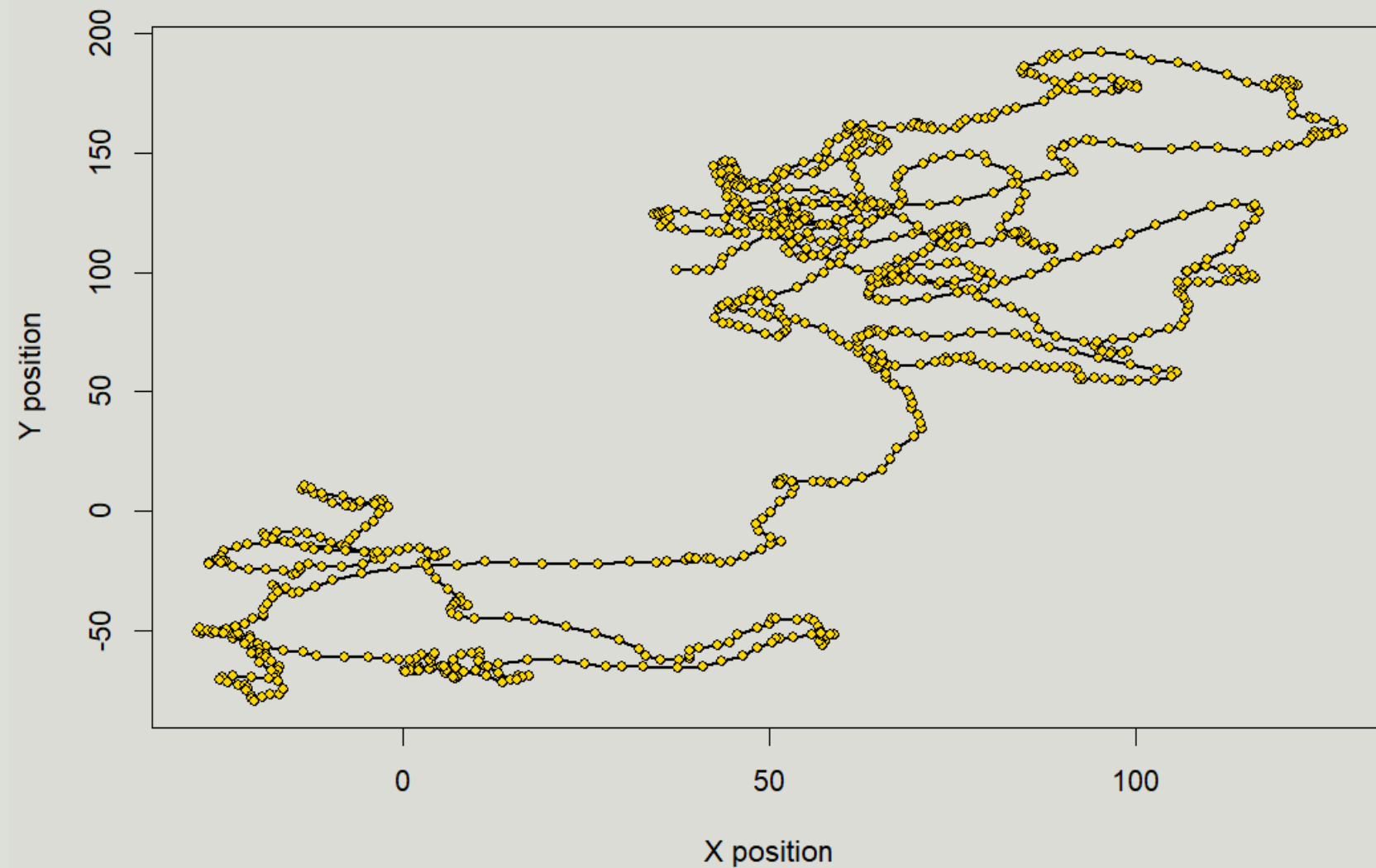
Observaciones
independientes



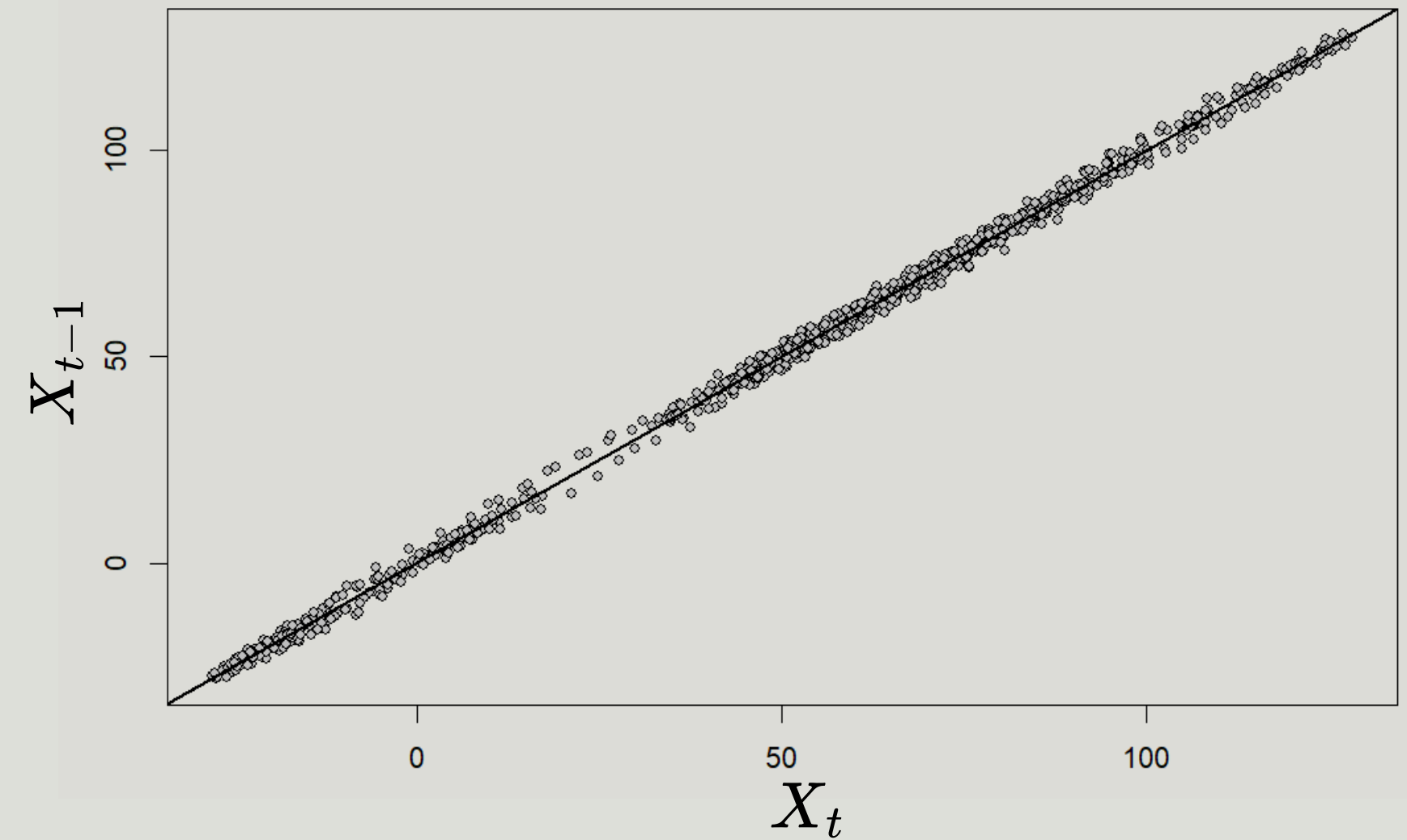
{ EE pequeños
IC angostos
p << 0.0001

Autocorrelación

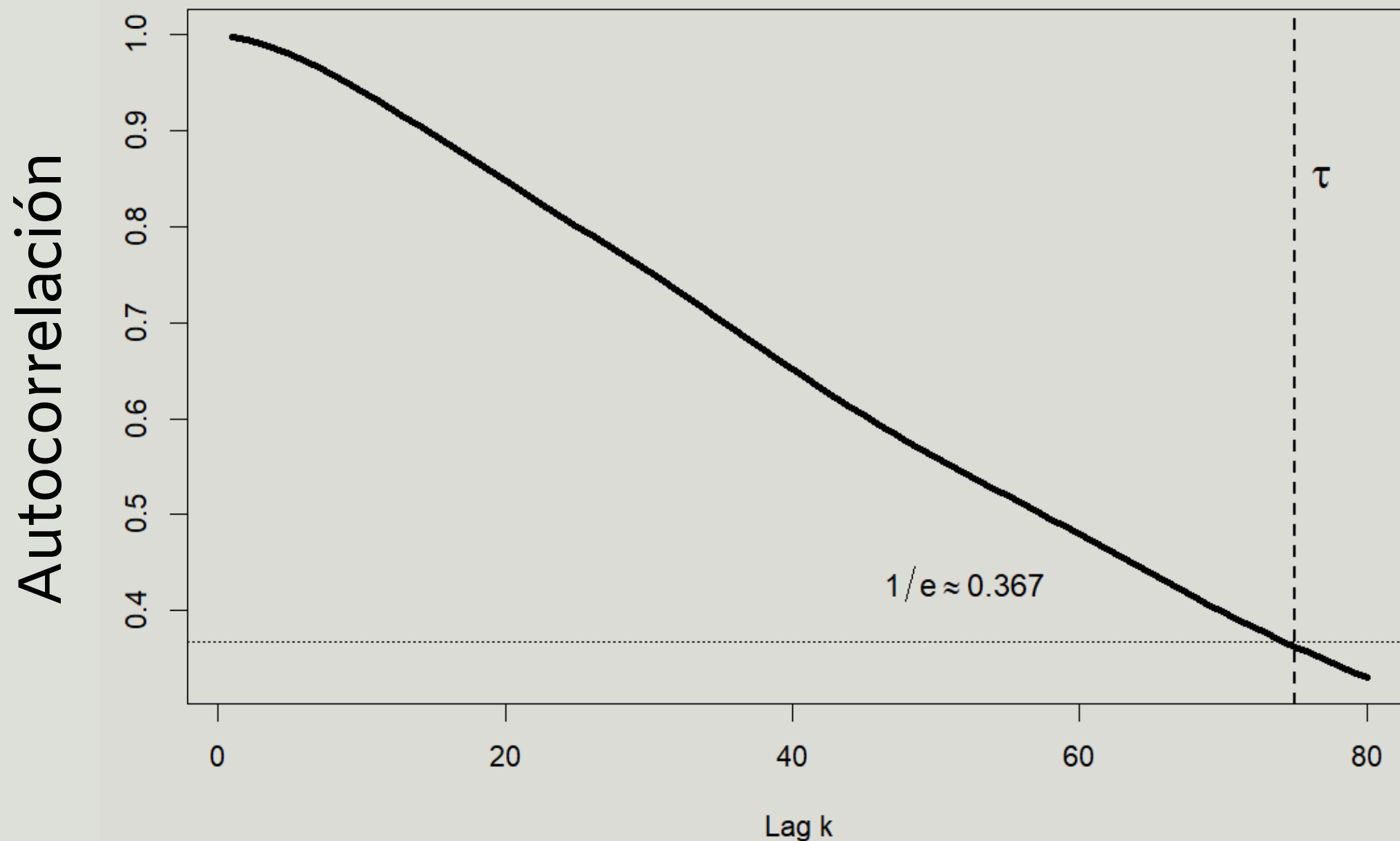
Observaciones arrastran memoria



$$\text{Cor}(X_t, X_{t+1}) \approx 1$$



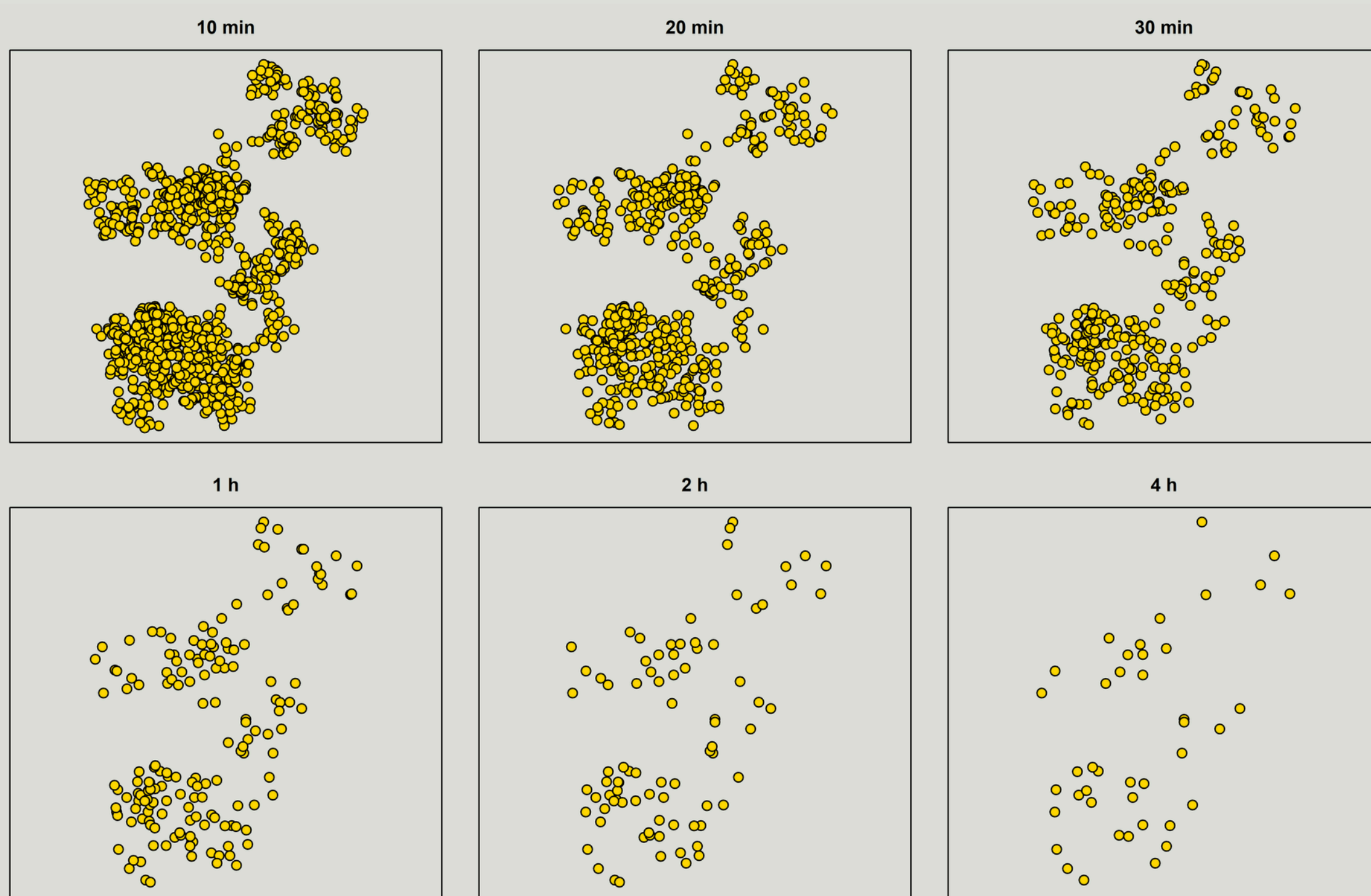
Mientras más lejos en el tiempo, mejor..
¿Cuán lejos en el tiempo?



$$\tau = \min \left\{ k : \rho(k) < \frac{1}{e} \right\}$$

$$\rho(k) = \text{Cor}(X_t, X_{t+k})$$

Tamaño muestral $\neq$ Tamaño efectivo



Mejor caso
posible

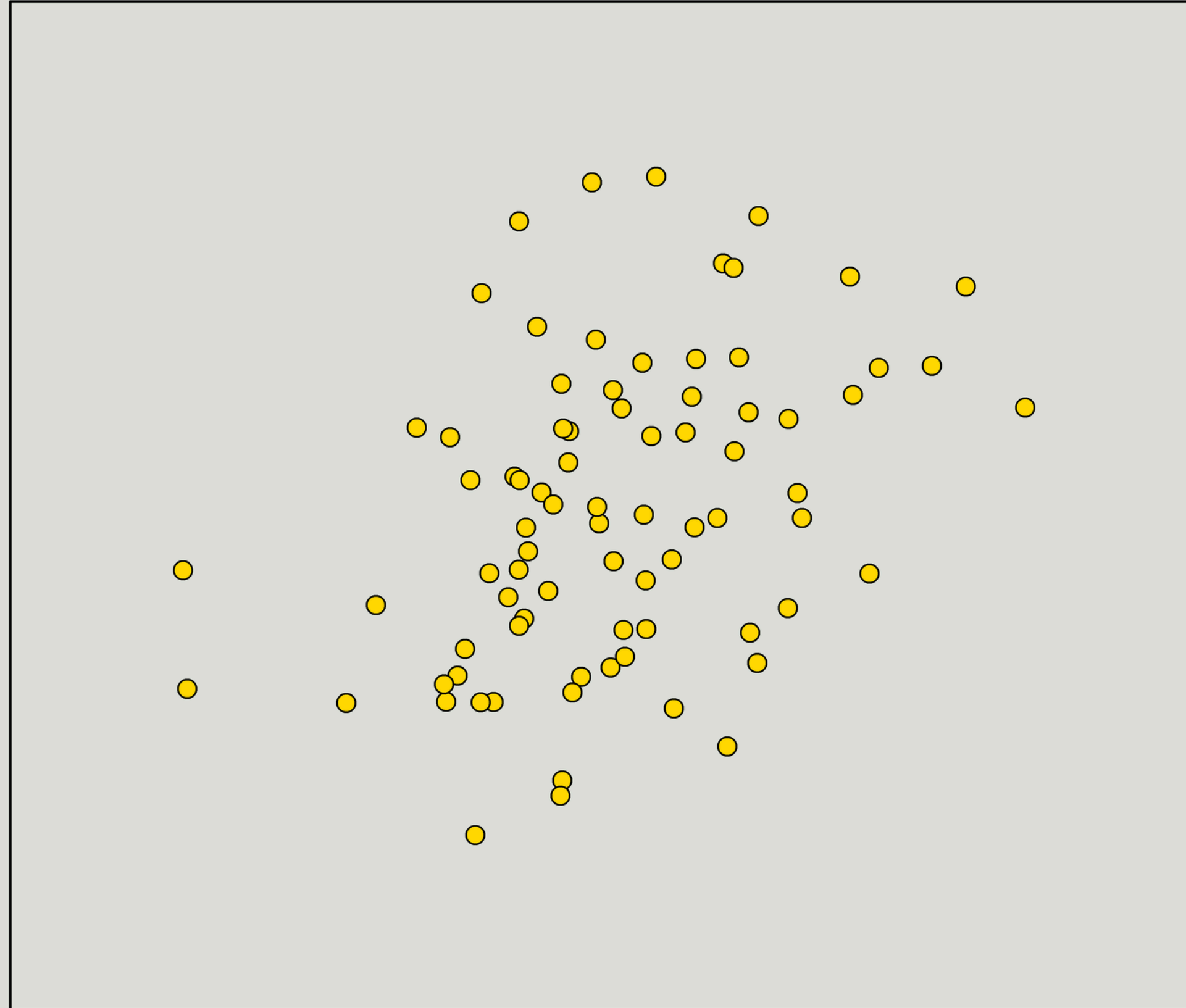
$$\Delta t = \tau$$



$$\eta_{eff} = \eta_{muestral}$$

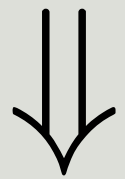
Por dónde empezamos...

Pts Usados

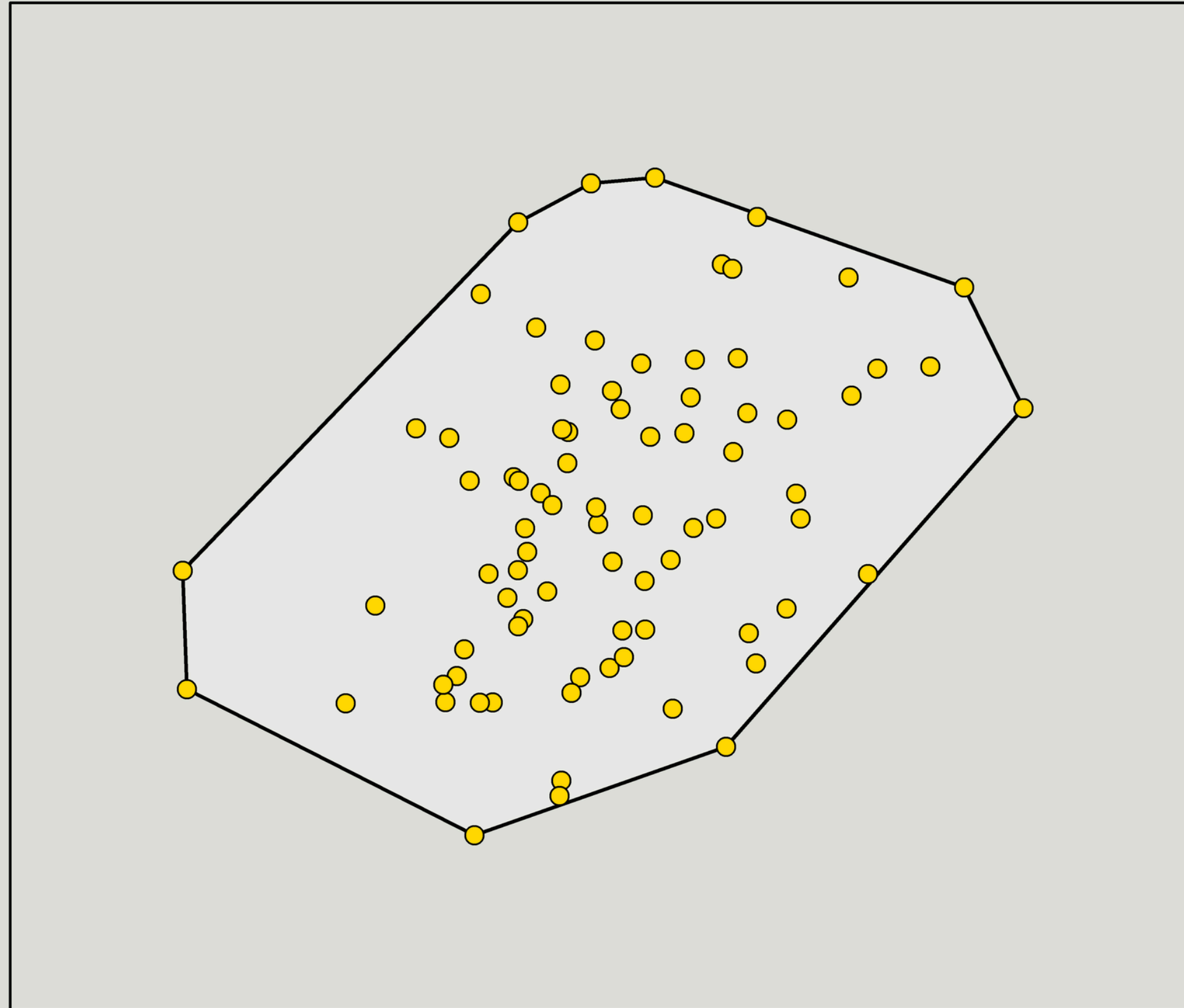


Por dónde empezamos...

Pts Usados

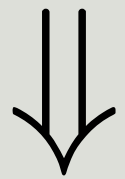


MCP 100%

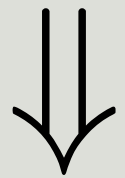


Por dónde empezamos...

Pts Usados

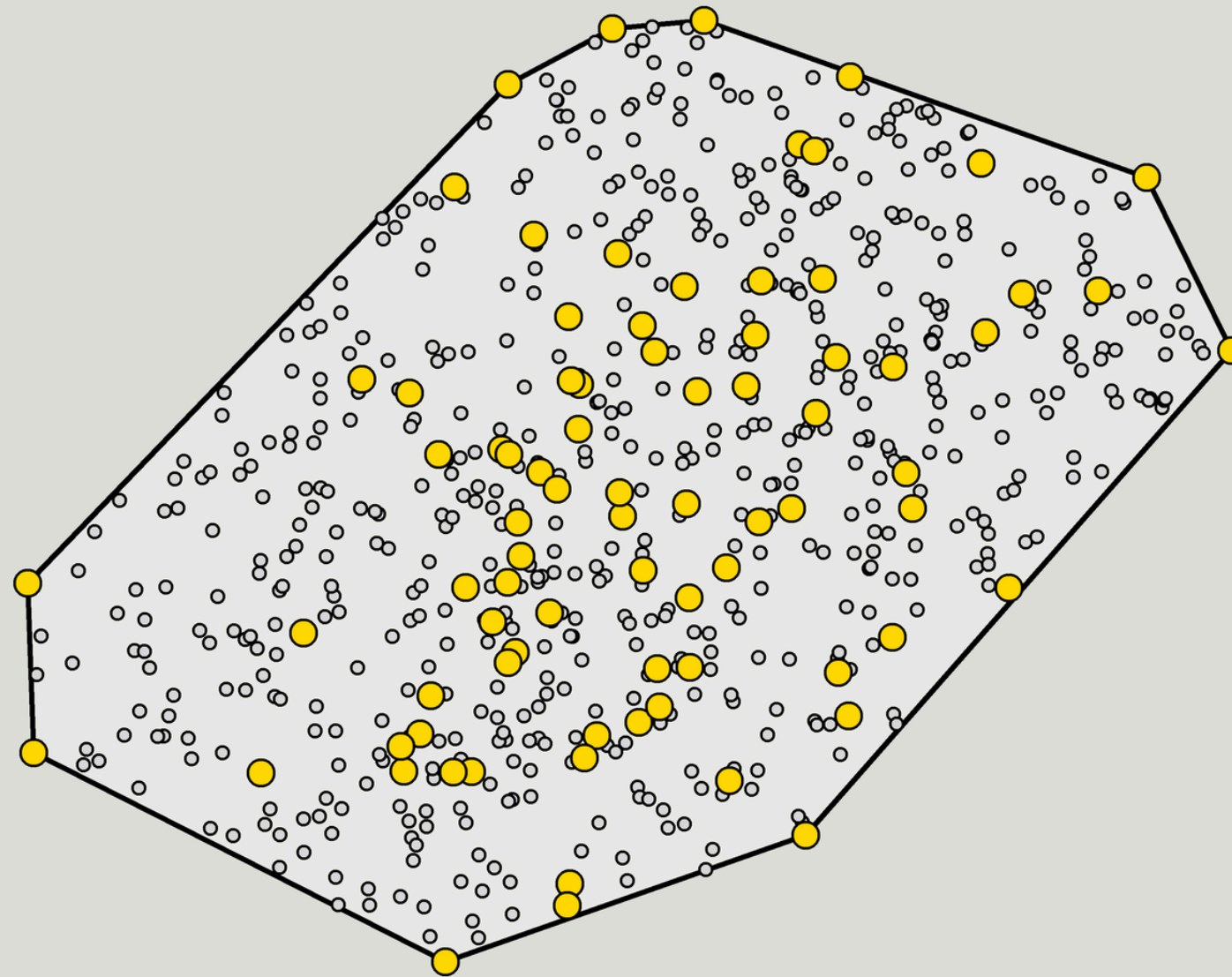


MCP 100%



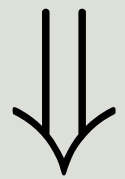
Pts
Disponibles

Azar

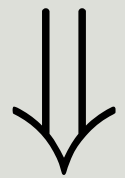


Por dónde empezamos...

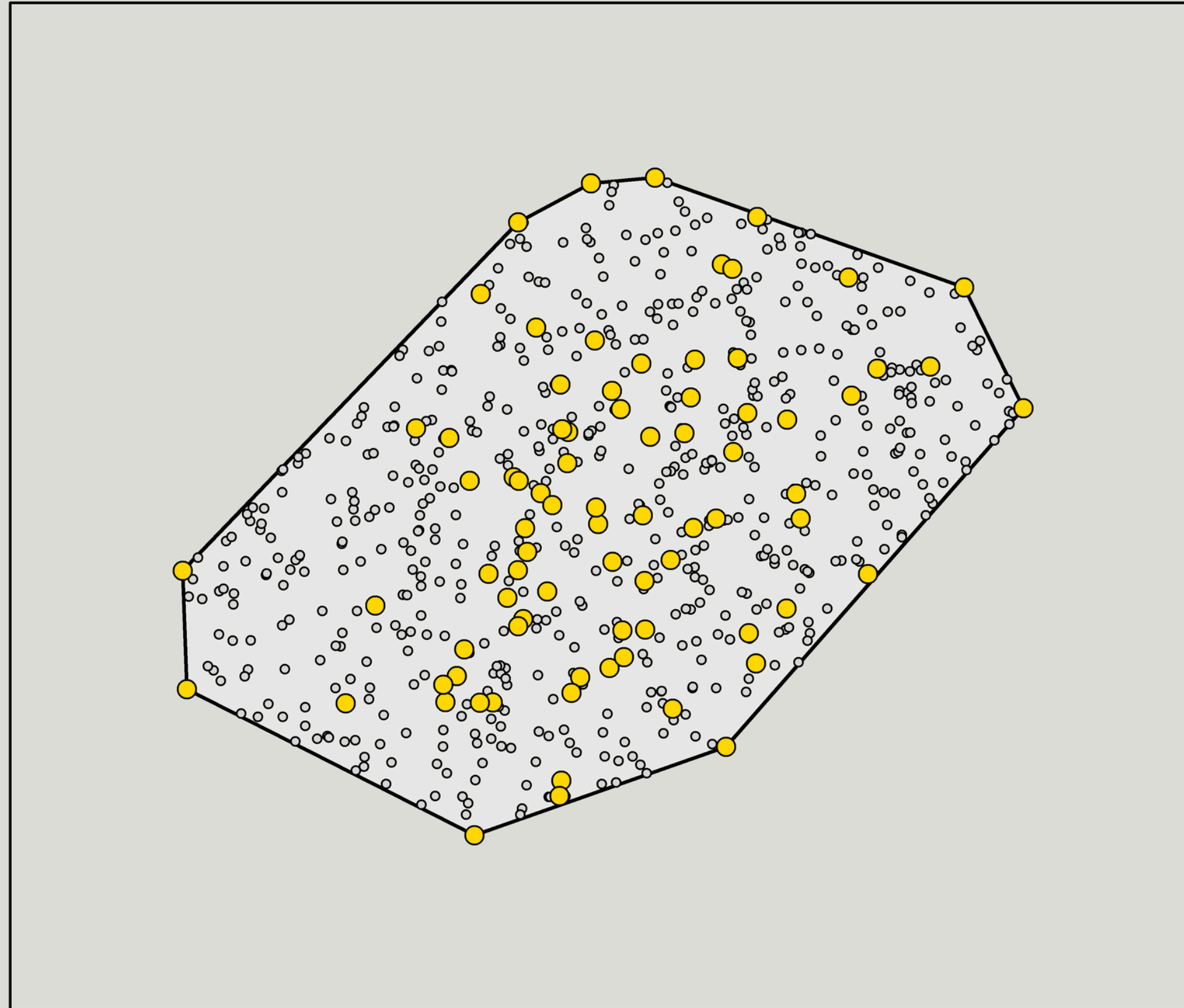
Pts Usados



MCP 100%

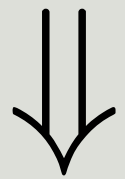


Pts
Disponibles

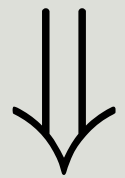


Por dónde empezamos...

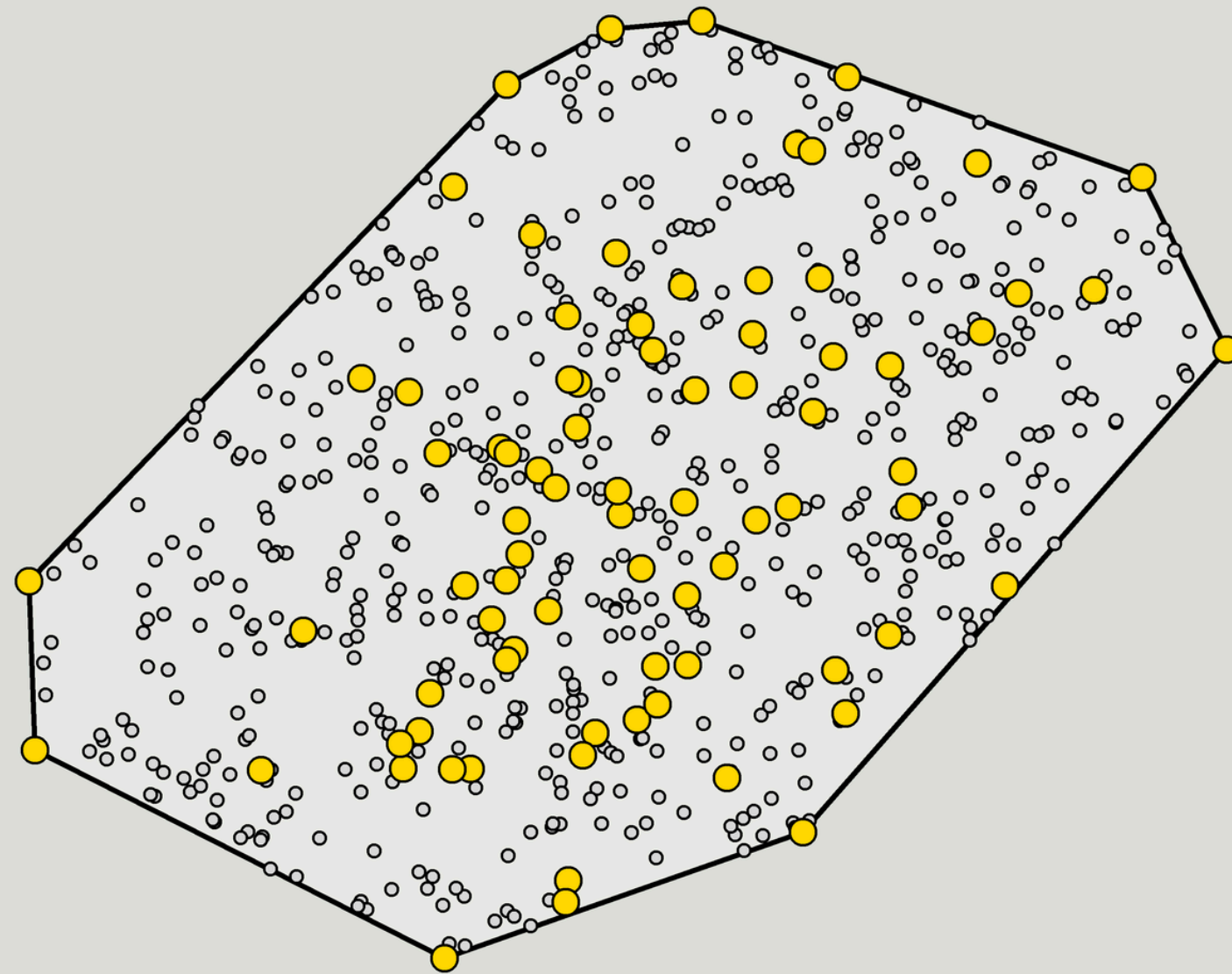
Pts Usados



MCP 100%

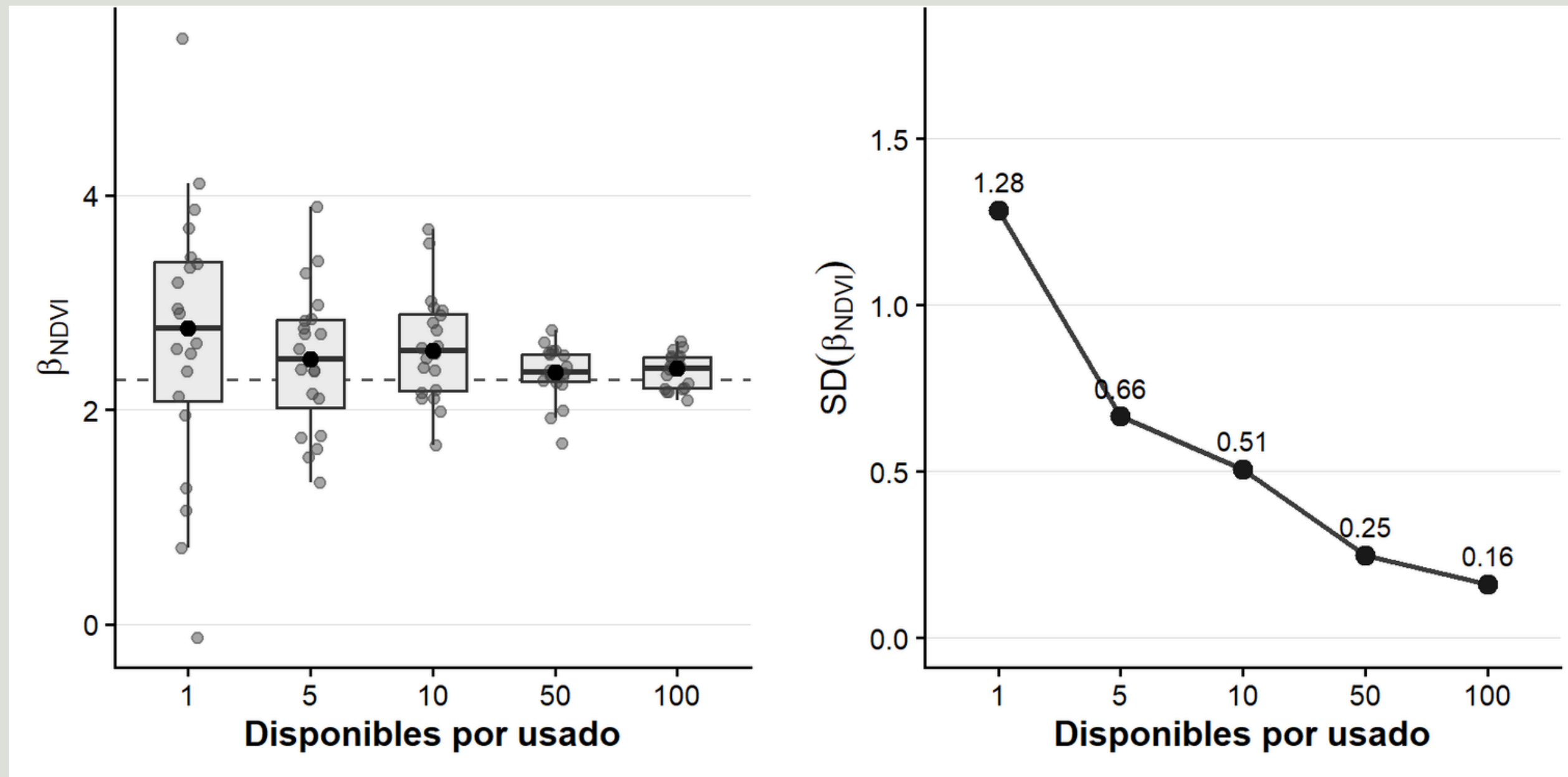


Pts
Disponibles

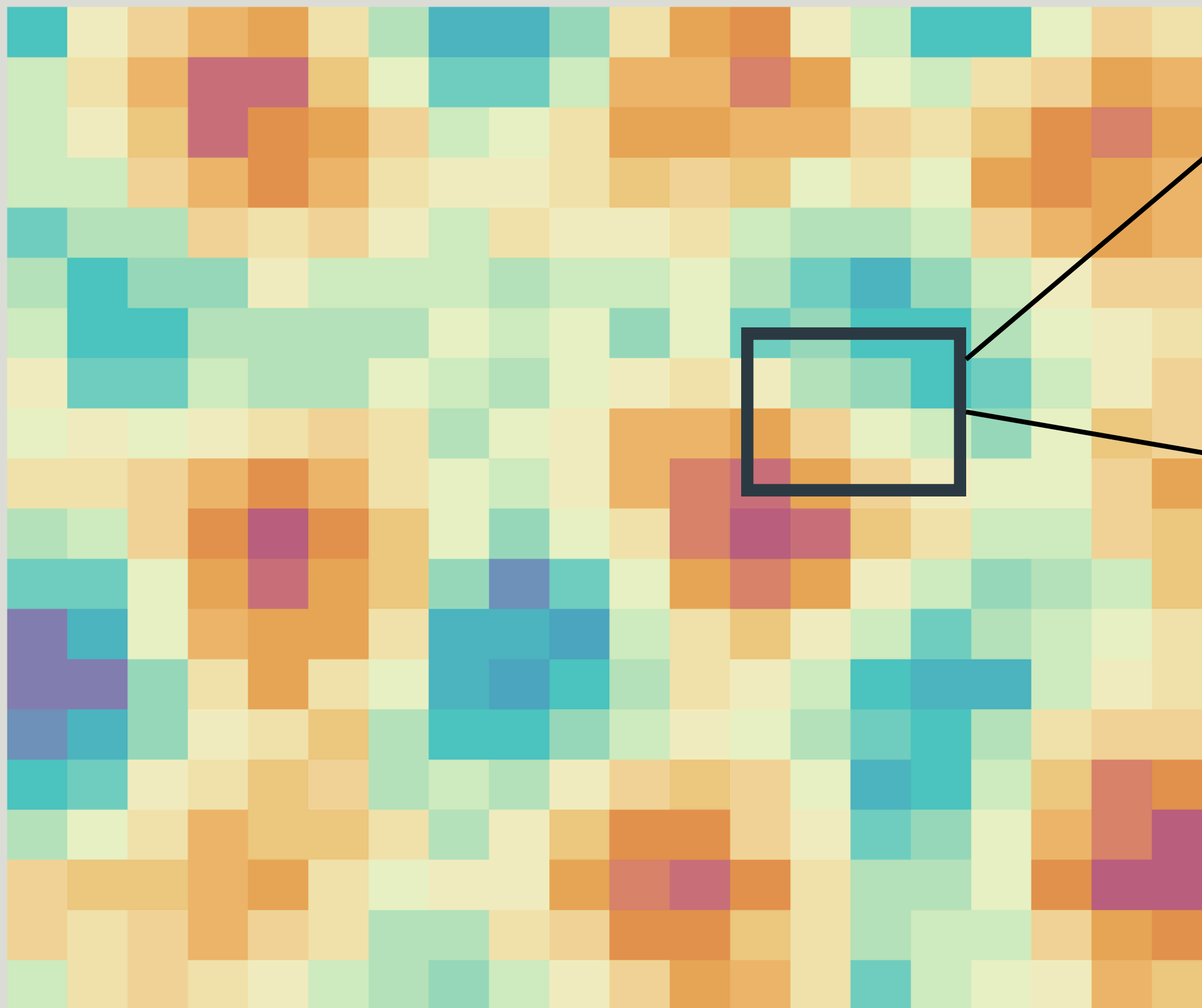


Análisis de Sensibilidad

Número de puntos disponibles que reduzca la variación entre estimadores

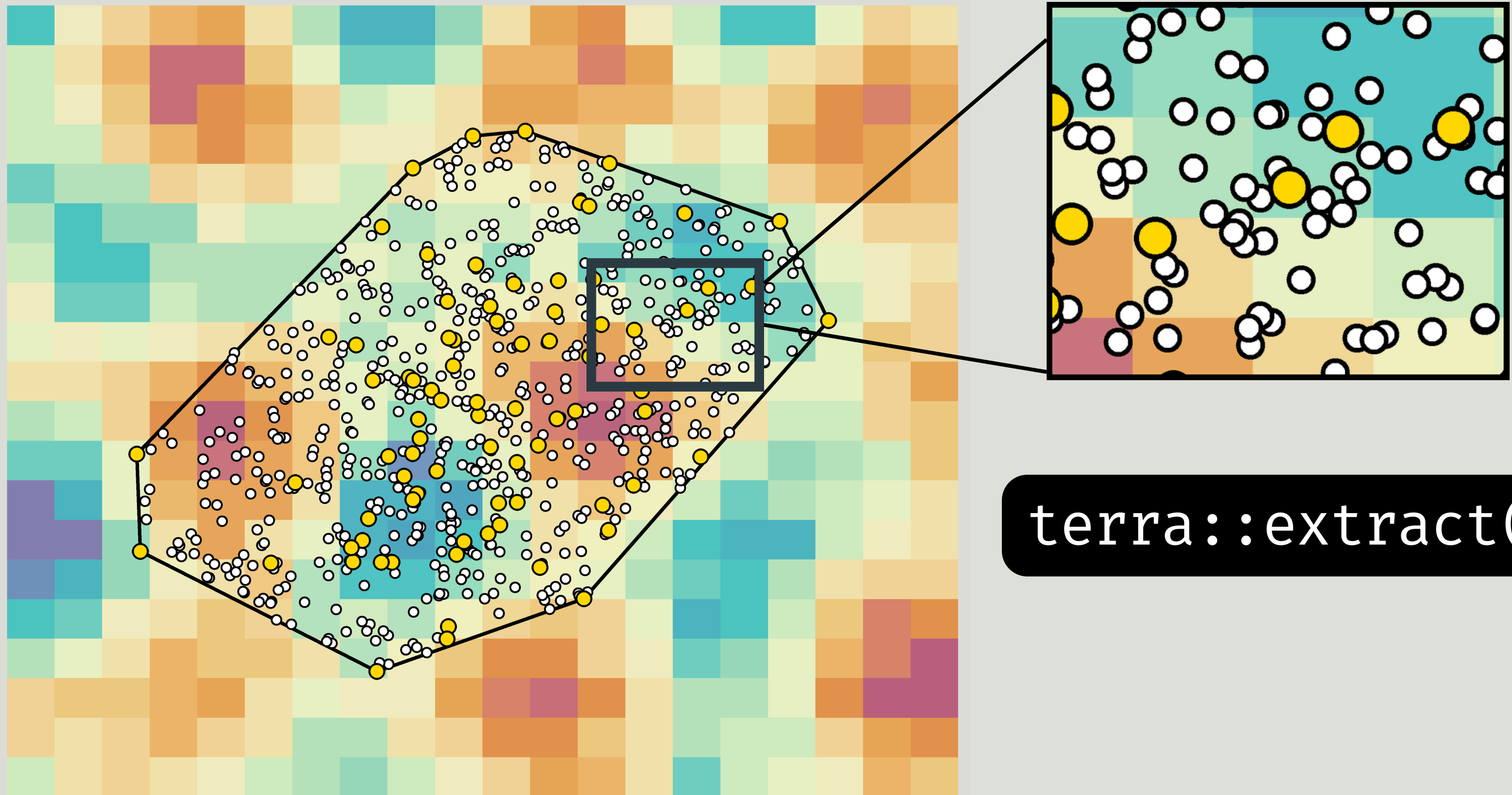


Covariables



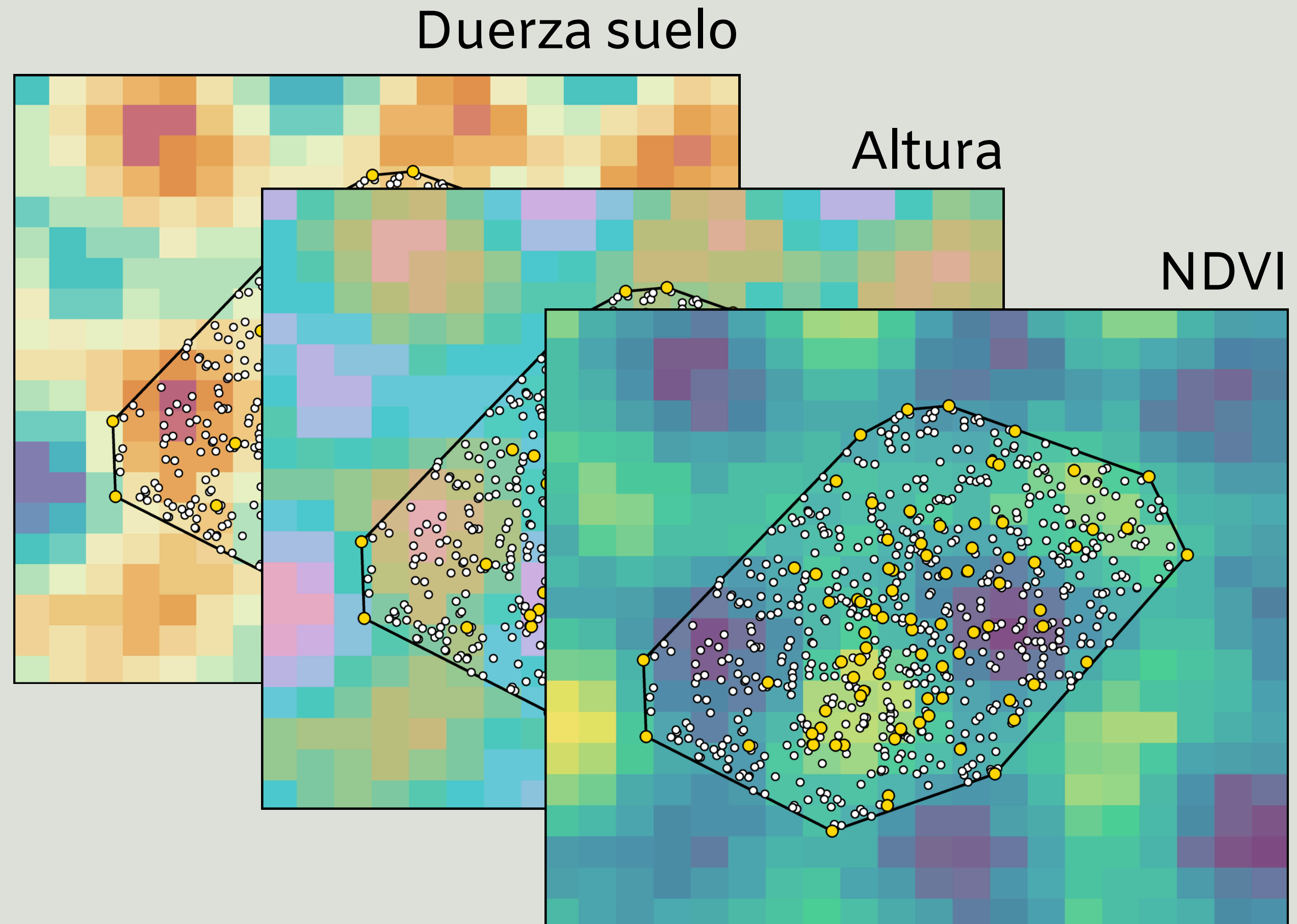
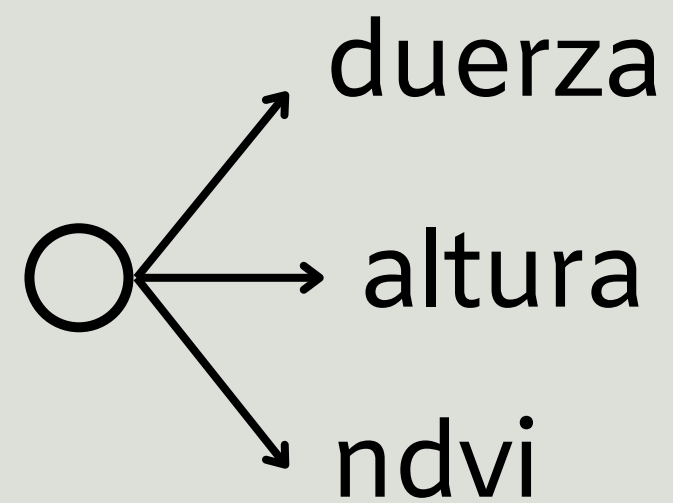
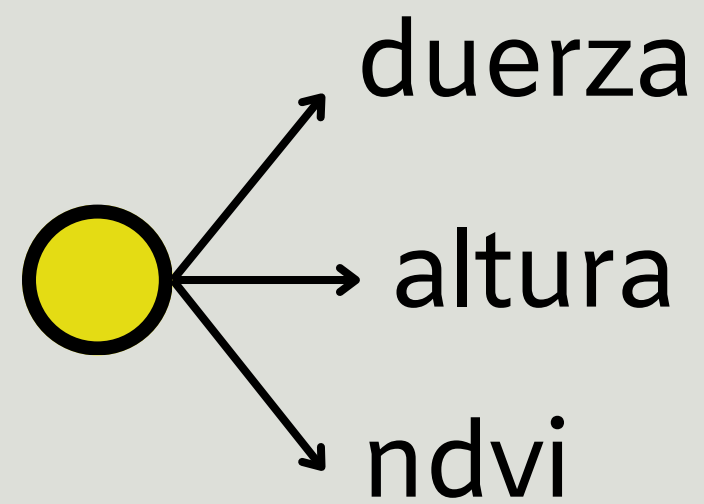
1	2	2	3
0	1	3	6
2	1	5	7
0	0	4	8

Covariables



`terra::extract()`

Capas de Rasters



Se ajusta RSF

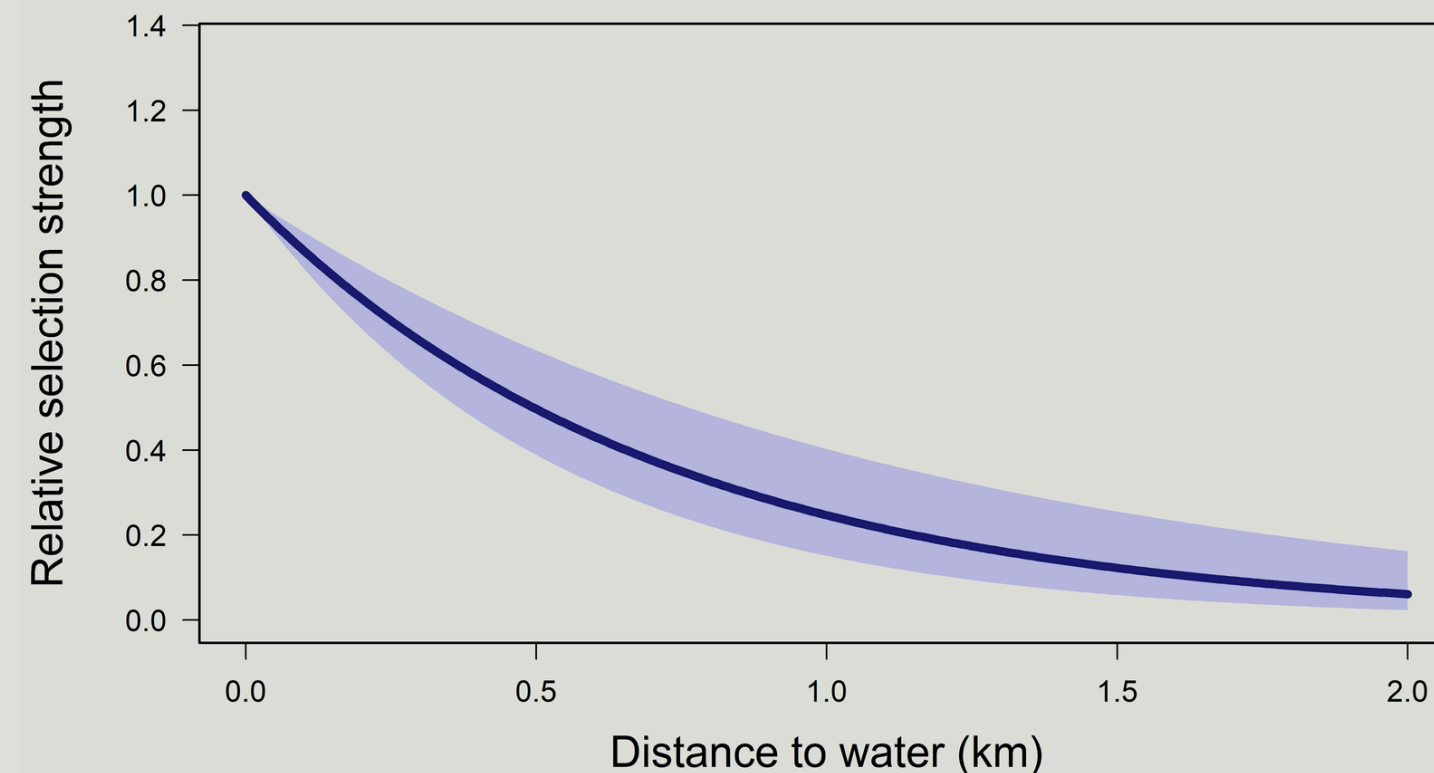
```
m_rsf <- glm(used ~ NDVI,  
             data = rsf_dat,  
             family = binomial,  
             weights = w)
```

$$\begin{aligned} \bullet & \left\{ w = 1 \right. \\ \circ & \left\{ w = \frac{\text{Total usados}}{\text{Total Disponibles}} \right. \end{aligned}$$

Se interpreta

```
summary(m_rsf)
```

```
exp(coef(m_rsf))
```



Se puede extrapolar

